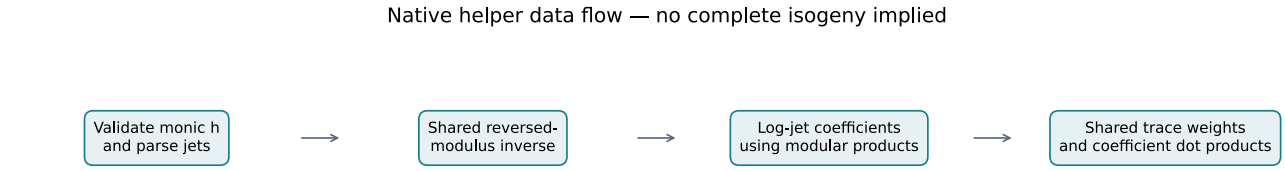
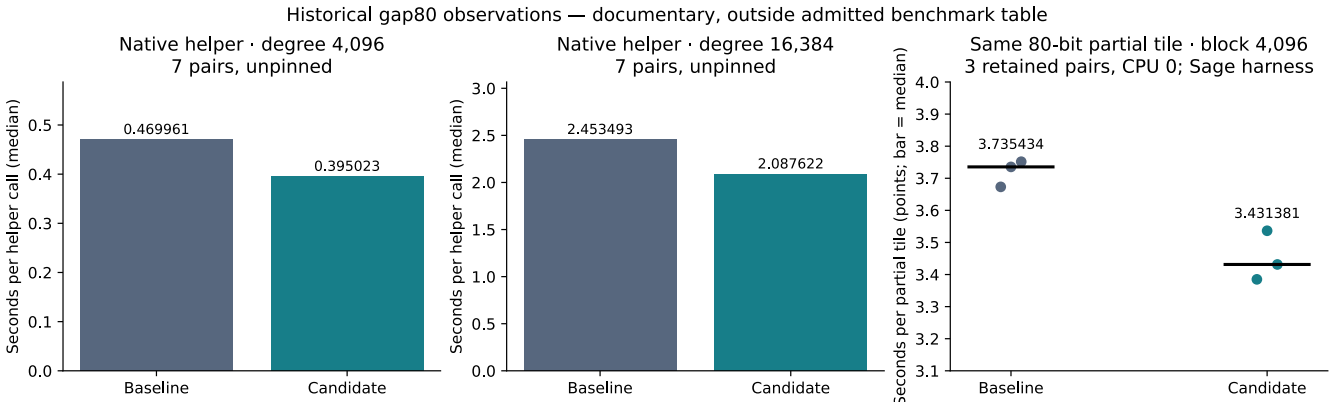
Native helper and 80-bit source/kernel

The native helper shares the trace functional and the modulus reciprocal. It passed the 96-case direct-root oracle and 96 rejection controls across three fields. The independent PARI checks verify the frozen source/kernel arithmetic and a replayable 202-bit field-prime certificate.



Retained observations, not admitted performance rows



The saved ratio of baseline/candidate medians is 1.189706 at native polynomial degree 4,096 and 1.175257 at degree 16,384 (seven pairs each, unpinned). On the same 80-bit partial tile it is 1.088610 (three retained CPU-0 pairs), an arithmetic 8.139711% reduction in that interval. These are historical diagnostics. The run report documents Python/Sage tile provenance, all exclusions, host limits and missing uncertainty/replay. No result is promoted into the admitted benchmark or IC/rho tables.

Exact candidate and open construction

The prime degree is 72984542927631840968704 1 (80 bits). The source is $y^2 = x^3 - 140x + 784$ over the 202-bit prime in `../parameters.json`. The ordinary CM source has $j = -3375$ and maximal order of discriminant -7; the inert prime-degree kernel gives the structural conductor argument. Native arithmetic and independent PARI verify the rational kernel and 41-bit subgroup, plus a trace-sign witness. This is not a formal proof assistant certificate or an explicit target-curve computation.

The explicit quotient equation and complete non-kernel images remain uncomputed. At block 4,096 the current bounded method needs 10,875,544,558,802,641 rectangular tiles plus 722,432,689,208 leftover roots. The next construction must complete the whole map with recorded native costs. Matched endpoint trials depend on that destination and transported base. No 80-bit evaluated crossing or largest-known record is established. The published 263-bit comparison does not establish one either.

Evidence, admission and follow-up

The source-linked study README, unchanged summary, parameters, field-prime certificate, frozen archive and checksum are one directory above. Native C source and portable build/verification entry points are in `../native/`. Historical Python remains archived evidence, not an active execution path.

The existing obligation is tracked in `problems/conductor-gap-upper-floor-and-cost.md`. Documentary copies live in `benchmarks/20261031-gap80-native-diagnostic/`, outside the admitted table under the validation gate. Canonical leaderboard, IC/rho progress, curve registry and browser graphs remain unchanged: no admitted comparison, completed crossing or registered curve promotion is supplied. References and the complete retained measurement flow are in the README/run report.